

Aditya T. Vadlamani

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EDUCATION

The Ohio State University

Columbus, OH

PhD in Computer Science & Engineering

Aug. 2023 – Present

- **Research focus:** Graph Machine Learning and Conformal Prediction

M.S. in Computer Science & Engineering

Jan. 2022 – May 2023

B.S. in Computer Science & Engineering, Mathematics, Magna Cum Laude

Aug. 2018 – May 2022

EXPERIENCE

Machine Learning Engineer Intern

May 2026 – Aug. 2026

Pinterest

Remote

- Introduced a new architecture and training objectives of OmniSage, Pinterest's in-house representation learning framework, to better leverage user engagement signals
- Improved key offline evaluation metrics by up to 5% through the new architecture and objectives

Software Engineer Intern

May 2025 – Aug. 2025

Google

Sunnyvale, CA

- Designed and implemented a TF-GNN-based graph neural network to rank code tests most likely to fail when machine learning libraries are modified across Google's codebase
- Built a large-scale data pipeline using an internal Flume-based framework to construct code-test dependency graphs from Google's codebase for model training

Software Engineer Intern

May 2023 – Aug. 2023

Cruise LLC

Sunnyvale, CA

- Designed and implemented a prototype encoder model to detect invalid autonomous vehicle simulation scenarios
- Reduced manual review and accelerated software releases by filtering low-quality simulation outputs

Software Development Engineer Intern

May 2022 – Aug. 2022

Amazon Web Services

Cupertino, CA

- Prototyped a scalable AWS environment for functional network verification of routing policies.
- Implemented syntactic validation for routing policy attributes, preventing numeric and regular-expression errors that could otherwise trigger unnecessary router redeployments.

RESEARCH EXPERIENCE

Graduate Student Researcher

Aug. 2022 – Present

The Ohio State University

Columbus, OH

- Developed a framework for algorithmically fair conformal prediction and extended it for federated learning.
- Benchmarked and analyzed conformal prediction methods for transductive node classification, identifying practical considerations for uncertainty quantification in GNNs.
- Developed scalable training methods for heterogeneous GNNs using sampling and ensembling techniques.
- Extending conformal prediction to generative models, including steering diffusion models and providing formal safety guarantees for LLMs for auditing.
- Administered multi-GPU high-performance computing infrastructure supporting large-scale machine learning and data-intensive research workloads.

SELECTED PUBLICATIONS

A Generic Framework for Conformal Fairness, ICLR 2025. Acceptance Rate: 31.75%

Conformal Prediction: A Theoretical Note and Benchmarking Transductive Node Classification in Graphs, TMLR 2025

BLB-HGNN: Bag of Little Bootstraps for Training Heterogeneous GNNs, ICDM 2025. Acceptance Rate: 22.42%

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, SQL, MATLAB, JavaScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, HuggingFace Transformers, NumPy, Pandas

Graph Machine Learning: PyTorch Geometric, DGL, TF-GNN, NetworkX

MLOps & Experiment Tracking: Docker, Weights & Biases, MLflow

Distributed Systems & HPC: Ray, SLURM, CUDA, Apache Spark, Apache Flume

Cloud Platforms & DevOps: AWS, Google Cloud Platform, Terraform, Git